

7. Integration Techniques

7.4. Partial Fraction Decomposition

7.4.3. Integrals Needing Partial Fraction Decomposition (Continued)

7.4b Learning Objectives

- I can use partial fraction decomposition to help me solve integrals.

Example 7.4.1

Compute the following antiderivative

$$\int \frac{3e^{2x} - 5e^{3x}}{(3e^x - 1)(e^x - 1)^2} dx$$

Note

If you feel like you need to review exponent rules do that, note

$$e^{3x} = e^{2x}e^x$$

for example.

Solution

First, it looks like e^x is everywhere, so we will do a quick u -substitution with $u = e^x$. Then $du = e^x dx$

$$\begin{aligned} \int \frac{3e^{2x} - 5e^{3x}}{(3e^x - 1)(e^x - 1)^2} dx &= \int \frac{3e^x - 5e^{2x}}{(3e^x - 1)(e^x - 1)^2} e^x dx \\ &= \int \frac{3u - 5u^2}{(3u - 1)(u - 1)^2} du \end{aligned}$$

Now, this integral needs a partial fraction decomposition. The denominator is already factored into all linear terms so the form of the partial fractions is

$$\frac{3u - 5u^2}{(3u - 1)(u - 1)^2} = \frac{A}{3u - 1} + \frac{B}{u - 1} + \frac{C}{(u - 1)^2}$$

Multiplying everything by the common denominator

$$\begin{aligned} \frac{3u - 5u^2}{(3u - 1)(u - 1)^2} ((3u - 1)(u - 1)^2) &= \left(\frac{A}{3u - 1} + \frac{B}{u - 1} + \frac{C}{(u - 1)^2} \right) ((3u - 1)(u - 1)^2) \\ 3u - 5u^2 &= A(u - 1)^2 + B(u - 1)(3u - 1) + C(3u - 1) \end{aligned}$$

We can find A, C easily by plugging in values $u = 1, u = \frac{1}{3}$ but will have to deal with B another way. Then, for $u = 1$

$$3(1) - 5(1)^2 = A((1) - 1)^2 + B((1) - 1)(3(1) - 1) + C(3(1) - 1)$$

$$-2 = 2C$$

so $C = -1$. Then

$$3\left(\frac{1}{3}\right) - 5\left(\frac{1}{3}\right)^2 = A\left(\left(\frac{1}{3}\right) - 1\right)^2 + B\left(\left(\frac{1}{3}\right) - 1\right)\left(3\left(\frac{1}{3}\right) - 1\right) + C\left(3\left(\frac{1}{3}\right) - 1\right)$$

$$\frac{4}{9} = \frac{4}{9}A$$

so $A = 1$. Then plugging in A, C we have

$$3u - 5u^2 = (u - 1)^2 + B(u - 1)(3u - 1) - (3u - 1)$$

Since we only need to find one of the coefficients still, we can plug in any value for u we haven't used here and solve for B . Let's do $u = -1$. Then,

$$3(-1) - 5(-1)^2 = ((-1) - 1)^2 + B((-1) - 1)(3(-1) - 1) - (3(-1) - 1)$$

$$-8 = 4 + B(-2)(-4) - (-4)$$

$$-8 = 4 + 8B + 4$$

$$-16 = 8B$$

so $B = -2$. Then, the integrand becomes

$$\int \frac{3e^{2x} - 5e^{3x}}{(3e^x - 1)(e^x - 1)^2} dx = \int \frac{A}{3u - 1} + \frac{B}{u - 1} + \frac{C}{(u - 1)^2} du$$

$$= \int \frac{1}{3u - 1} - 2\frac{1}{u - 1} - \frac{1}{(u - 1)^2} du$$

$$= \int \frac{1}{3u - 1} du - 2 \int \frac{1}{u - 1} du - \int \frac{1}{(u - 1)^2} du$$

We can solve each of these new integrals with a small substitution. First, let $w = 3u - 1$ then $dw = 3 du$ and so

$$\int \frac{1}{3u - 1} du = \frac{1}{3} \int \frac{1}{w} dw$$

$$= \frac{1}{3} \ln|w|$$

$$= \frac{1}{3} \ln|3u - 1|$$

$$= \frac{1}{3} \ln|3e^x - 1|,$$

where we might as well get back to x while we are here. Then with $y = u - 1$ so $dy = du$ the second integral becomes

$$\begin{aligned}
 -2 \int \frac{1}{u-1} du &= -2 \int \frac{1}{y} dy \\
 &= -2 \ln|y| \\
 &= -2 \ln|u-1| \\
 &= -2 \ln|e^x - 1|.
 \end{aligned}$$

Lastly, with $z = u - 1$ so $dz = du$ the last integral becomes

$$\begin{aligned}
 - \int \frac{1}{(u-1)^2} &= - \int z^{-2} dz \\
 &= -\frac{1}{-1} z^{-1} \\
 &= \frac{1}{z} \\
 &= \frac{1}{u-1} \\
 &= \frac{1}{e^x - 1}
 \end{aligned}$$

Altogether our final answer is

$$\int \frac{3e^{2x} - 5e^{3x}}{(3e^x - 1)(e^x - 1)^2} dx = \frac{1}{3} \ln|3e^x - 1| - 2 \ln|e^x - 1| + \frac{1}{e^x - 1} + C.$$

There are no extra considerations needed for the bounds of these integrals. If you end up needing a substitution, change the bounds accordingly and so on.

Exercise 7.4.1

Evaluate

$$\int_3^6 \frac{2x^3 - x^2 + 4x}{x^4 + 4x^2}$$

(You do not need to evaluate any complicated functions at the end.)

Example 7.4.2

Compute the following antiderivative

$$\int \frac{x^4 + 2x^2 + 1}{x - 1} dx$$

Solution

First, $\deg \text{ numerator} \geq \deg \text{ denominator}$ so we have to do polynomial long division first.

$$\begin{array}{r}
 x^3 + x^2 + 3x + 3 \\
 x-1 \overline{) \begin{array}{r} x^4 + 2x^2 + 1 \\ -(x^4 - x^3) \\ \hline x^3 + 2x^2 + 1 \\ -(x^3 - x^2) \\ \hline 3x^2 + 1 \\ -(3x^2 - 3x) \\ \hline 3x + 1 \\ -(3x - 3) \\ \hline 4 \end{array}}
 \end{array}$$

So,

$$\frac{x^4 + 2x^2 + 1}{x - 1} = (x^3 + x^2 + 3x + 3) + \frac{4}{x - 1}$$

Then

$$\begin{aligned}
 \int \frac{x^4 + 2x^2 + 1}{x - 1} dx &= \int (x^3 + x^2 + 3x + 3) dx + \int \frac{4}{x - 1} dx \\
 &= \frac{x^4}{4} + \frac{x^3}{3} + \frac{3}{2}x^2 + 3x + 4 \ln|x - 1| + C
 \end{aligned}$$

7.4b Section Summary:

- We solved a lot of integrals using partial fraction decomposition.